Module Assignment & Exam Outlines

Module code and Name	**ELEC70126 - Internet of Things and Applications**
Module Leader	**Hamed Haddadi**
Assessment name & weighting	**Coursework, 100%**
Assessment date	**2pm Weds 25/03/2026 (submit once only) (optional half-page work-plan to be submitted by 11/02/2026 for feedback, this piece is not part of the final marks)**

Summary description and objectives for the assessment

Part 1: Sensing (50%)

For at least a week's worth of continuous data from at least 2 time-series data sources of different nature (e.g., ambient data from sensors, mobile sensors, social media, smart city APIs, etc), set up a platform to automatically:

1. Acquire data from a custom prototype sensor or a public data source
2. Establish a network communications and/or data storage to save the data
3. Justify the design choices, and discuss any data cleaning and/or filtering

Part 2: Analytics and interface (50%)

Use the data from part 1 and design and implement a prototype for:

1. A web App/phone app/other user interface/interaction form (wearable, smarthome, etc) to present the data and provide user engagement opportunities,
2. Perform Basic time series data analysis to provide added insight into the data,
3. Present and discuss opportunities Innovation, creativity, enterprise, scalability & complexity analysis within your ecosystem

Each individual must submit their own assignment independently.

Submission requirements (for assignments only)

- Submissions: 1 report to be submitted per student, combined PDF of up to 10 pages, one for part 1 and one for part 2 (up to 5 pages each), using the template on Blackboard

- Code & Data: link to accessible code (ideally Github, or publicly accessible Dropbox/OneDrive/etc folder), link to data repository

Learning outcomes (from module descriptor)	Assessment criteria

Having an understanding of the basic principles of Sensing, IoT, and digital signal processing	Correctness, appropriateness and depth of the technical acoustics and digital signal processing features within the developed work.
Haveingan understanding of the physical and social sensing systems and of the main principles of Internet of Things	Appropriateness and depth of the developed work in relation with the broader Internet of Things landscape.
Be able to integrate different sensor data software and hardware components, designing and creating actuations, analytics domains, products and/or complex interactions/experiences	Complexity and correctness of the integration between hardware and software components within the developed work. This criterion will be assessed considering also the use of software components in the design stage to predict and emulate the final product. If the work includes only software components, this criterion will be assessed considering also the graphic user interface of the developed work.
Have an experience of applying such knowledge to appropriate case studies	Appropriateness and depth of the solution found to solve the specific assigned tasks. Appropriateness of the planning and timeliness of the delivery. Appropriateness and depth of the reporting

Assessment criteria	Pass (D grade)	Excellent (A grade)	Weight (%)
Correctness, appropriateness and depth of the technical set up and digital signal processing features within the developed work.	Generally correct and appropriate features, but without technical depth.	The work shows features that are appropriate and correct considering the specific task, and that show a clear technical depth in terms of analysis and development.	30%
Appropriateness and depth of the developed work in relation with signal cross-correlation.	Basic but correct considerations of correlations of physical and social signals, without in-depth analysis.	Appropriate and extensive considerations of the time-series correlations and/or perception within the developed work	10%
Complexity and correctness of the integration between hardware and software components within the developed work.	Basic but appropriate integration between hardware and software components, aimed only at linking the two, and without	Correct and extensive integration of the hardware and software components.	10%

	considering the full potential of the interaction.		
If the work includes only software components, this criterion will be assessed considering also the graphic user interface of the developed work.	Basic but appropriate graphical user interface.	Appropriate graphical user interface which shows extensive work in the design, ease of use and access of the various component.	
Appropriateness and depth of the solution found to solve the specific scenario.	Basic but appropriate solution, correctly implemented.	Appropriate and correct solution, which demonstrates extensive depth in terms of analysis and technical features.	20%
Appropriateness of the planning and timeliness of the delivery.	Generally appropriate planning, with an overall timely delivery of the various components, without in-depth considerations about the various processes.	Appropriate and extensive planning, with timely delivery of the various components and in-depth considerations of the various stages of the design, development and delivery.	10%
Appropriateness and depth of the reporting	Appropriate report, containing required multimedia material, sufficiently well organised but without in-depth information about the various stages of the work. No in-depth considerations on the overall installation.	Appropriate and extensive report, containing all the required multimedia material, all appropriately organised and presented, and with in-depth information about the various stages of the design, planning, development and delivery of the work. In-depth considerations on the overall installation.	20%

Appendix: Guidance questions

For the various section of the report, think about the following questions:

• Executive summary:
o Have you outlined the background to the problem?
o Have you clarified the purpose of the report?
o Have you provided brief details of the approach you followed? o Have you summarised the important results and findings?
o Have you stated the major conclusion from your work?

• Design and implementation.
o Have you identified the problem you are addressing?
o Have you provided an overview of how you are addressing this problem?
o Have you provided detailed steps of what you did in a logical order?
o Have you described the methods you used to solve the problem and their purpose? o Have you identified the parameters for designing your system?
o Have you identified the metrics for evaluating the performance of your system?
o Have you used illustrations to give the reader a visual interpretation of your system?

• Results.
o Every result included must have a method set out in the design methodology.

o Likewise, every method should have some results.
o Have you used figures and tables to illustrate your points?
o Have you critically evaluated the quality and reliability of your results?
o Are your observations supported by evidence?
o Is the analysis and discussion relevant to the project and its context?
o Have you explained how well (or not) you have met the project's objectives?

• Trade-offs.
o Have you considered and quantified the cost (you will incur) versus the gain (you will

obtain) when you modify different parameters in your system? For example: ▪ Local vs. Cloud processing.
▪ Range vs. Power consumption (battery life).
▪ Throughput vs. Power consumption.

▪ Latency / Delay vs. Accuracy.
▪ Performance vs. Security.
▪ Generic vs. Bespoke hardware. ▪ Price vs. Functionality.

▪ Price vs. Usability.
▪ Price vs. Robustness / Reliability.
▪ Functionality vs. Usability.
▪ Functionality vs. Rapid development (time to market). ▪ Functionality vs. Backward compatibility.
▪ Flexibility vs. Accuracy vs. Interpretability.